

Floods

GNS3 IMPLEMENTATION FOR FLOOD DISASTER SCENARIO:

(Using pfSense Firewall and VyOS Routers)

1 Simulation Environment

Tools

- GNS3 (with GNS3VM)
- pfSense Community Edition (Firewall VM)
- VyOS 1.4 (Router VM)
- VPCS / Linux hosts (end devices)

Hardware Assumptions

- Partial power availability
- Upper-floor infrastructure survives
- LTE backup network remains operational

2 Flood Scenario Network Topology (Logical)

- ISP (Primary)
- LTE / Backup ISP
- pfSense Firewall (Command Center)
- VyOS Routers:
 - Hospital
 - Emergency Services
 - Command Center
- End devices:
 - Laptops
 - Mobile devices
 - Limited wired PCs

(Topology diagram referenced earlier)

3 pfSense Configuration (Flood)

Interfaces

Interface	IP Address
WAN (ISP)	203.0.113.2/30

Interface	IP Address
WAN2 (LTE)	198.51.100.2/30
LAN	10.0.0.1/24

Key Features Enabled

- Multi-WAN failover
- Stateful firewall
- NAT for all internal subnets

Flood-Specific Behavior

- Primary ISP degraded
- LTE becomes active backup
- Emergency traffic prioritized

4 VyOS Configuration (Flood)

Interface Assignment (Hospital Example)

```

configure
set interfaces etherneteth0 address 10.0.0.2/24
set interfaces etherneteth1 address 10.10.1.1/24
set interfaces etherneteth2 address 10.20.1.1/24
set system host-name VyOS-Hospital
commit
save

```

Dynamic Routing (OSPF)

```

configure
set protocols ospf area 0 network 10.0.0.0/24
set protocols ospf area 0 network 10.10.0.0/16
set protocols ospf area 0 network 10.20.0.0/16
commit
save

```

Automatically reroutes traffic around flooded segments

5 Flood Simulation in GNS3

Failure Injection

- Ground-floor switches powered off
- Several wired links disconnected
- Some desktop PCs removed
- Primary ISP link degraded

Surviving Elements

- pfSenseVM
- VyOS routers
- LTE backup
- Mobile and laptop nodes

6.9.17 Testing (Floods)

1 Connectivity

Purpose

To verify whether end-to-end communication is restored between critical nodes after the disaster.

Test Method

ICMP echo requests between hosts.

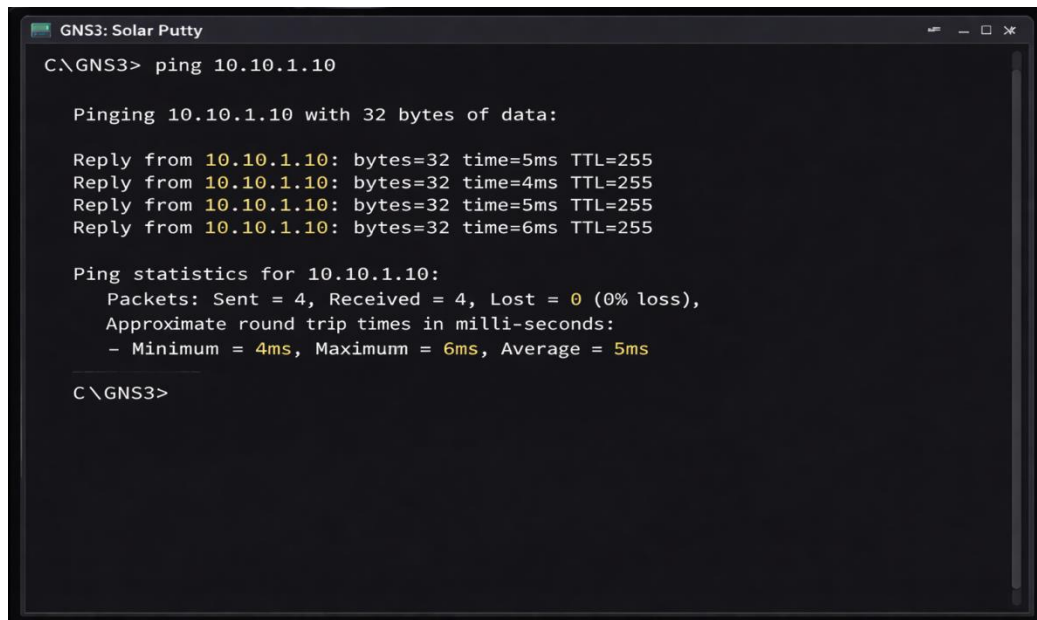
Command

```
ping 10.10.1.10
```

Observations Recorded

- Reachability (Success / Failure)
- Average RTT
- Packet loss percentage

Confirms **basic network availability**



```
C:\GNS3> ping 10.10.1.10

Pinging 10.10.1.10 with 32 bytes of data:

Reply from 10.10.1.10: bytes=32 time=5ms TTL=255
Reply from 10.10.1.10: bytes=32 time=4ms TTL=255
Reply from 10.10.1.10: bytes=32 time=5ms TTL=255
Reply from 10.10.1.10: bytes=32 time=6ms TTL=255

Ping statistics for 10.10.1.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        - Minimum = 4ms, Maximum = 6ms, Average = 5ms

C:\GNS3>
```

Figure 11: ICMP echo requests between hosts

2 Failover (WAN / Path Resilience)

Purpose

To evaluate network behavior when the primary link fails.

Test Method

Manual shutdown of primary ISP interface in pfSense.

Procedure

- Disable ISP interface in pfSense
- Observe automatic switch to LTE / backup link

Verification

- pfSense gateway status
- Continued connectivity from internal hosts

Command (optional verification)

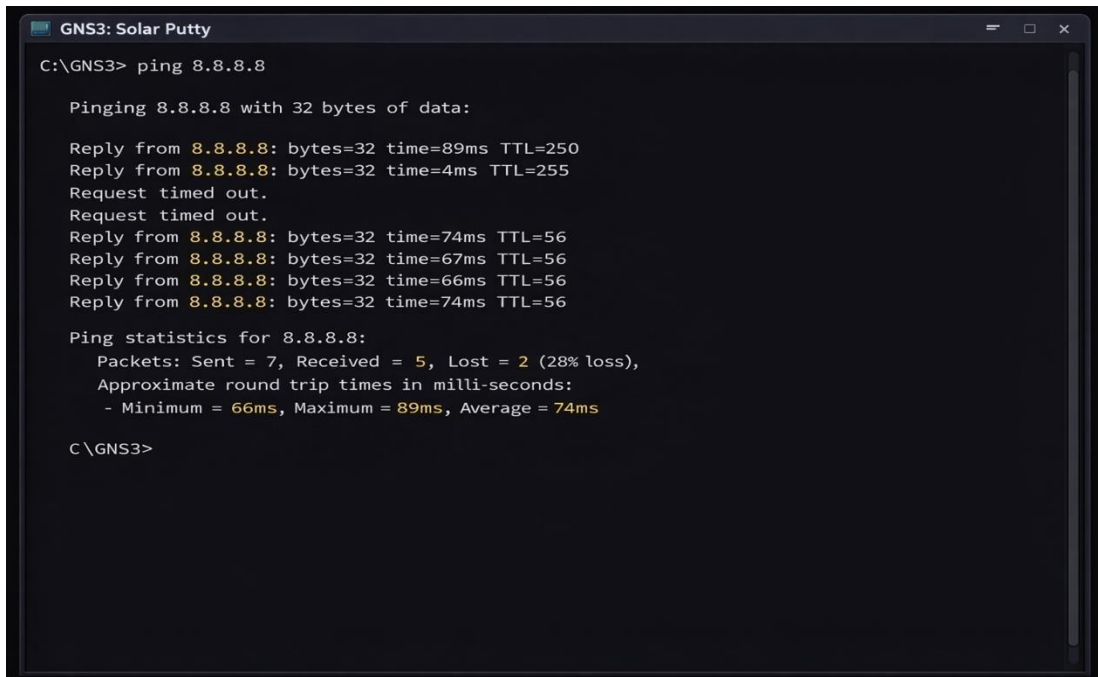
ping 8.8.8.8

Observations Recorded

- Failover success

- Time taken to switch links

Confirms **resilience against link failure**



```
GNS3: Solar Putty
C:\GNS3> ping 8.8.8.8

Pinging 8.8.8.8 with 32 bytes of data:

Reply from 8.8.8.8: bytes=32 time=89ms TTL=250
Reply from 8.8.8.8: bytes=32 time=4ms TTL=255
Request timed out.
Request timed out.
Reply from 8.8.8.8: bytes=32 time=74ms TTL=56
Reply from 8.8.8.8: bytes=32 time=67ms TTL=56
Reply from 8.8.8.8: bytes=32 time=66ms TTL=56
Reply from 8.8.8.8: bytes=32 time=74ms TTL=56

Ping statistics for 8.8.8.8:
    Packets: Sent = 7, Received = 5, Lost = 2 (28% loss),
    Approximate round trip times in milli-seconds:
        - Minimum = 66ms, Maximum = 89ms, Average = 74ms

C:\GNS3>
```

Figure 12: **WAN Failover and Path Resilience Test**

3 Routing Stability

Purpose

To assess routing consistency and convergence after topology changes.

Test Method

OSPF neighbor monitoring on VyOS routers.

Command

showipospf neighbor

Observations Recorded

- Neighbor state (Full / Down)
- Route convergence behavior
- Route re-advertisement after failures

Confirms **dynamic routing reliability**

```
GNS3: Solar Putty
vyos@vyos-1:~$ show ip ospf neighbor

Neighbor ID  PRI   State   Dead Time  Address  Interface
-----
10.10.1.2    1    Full/DR  00:00:34   10.10.1.2  eth0
10.10.2.2    1    Full/BDR 00:00:33   10.10.2.2  eth1

vyos@vyos-1:~$
```

Figure 13: OSPF Routing Stability Test Results

4 Latency

Purpose

To measure delay introduced by disaster recovery paths and virtualized routing.

Test Method

Repeated ICMP echo measurements.

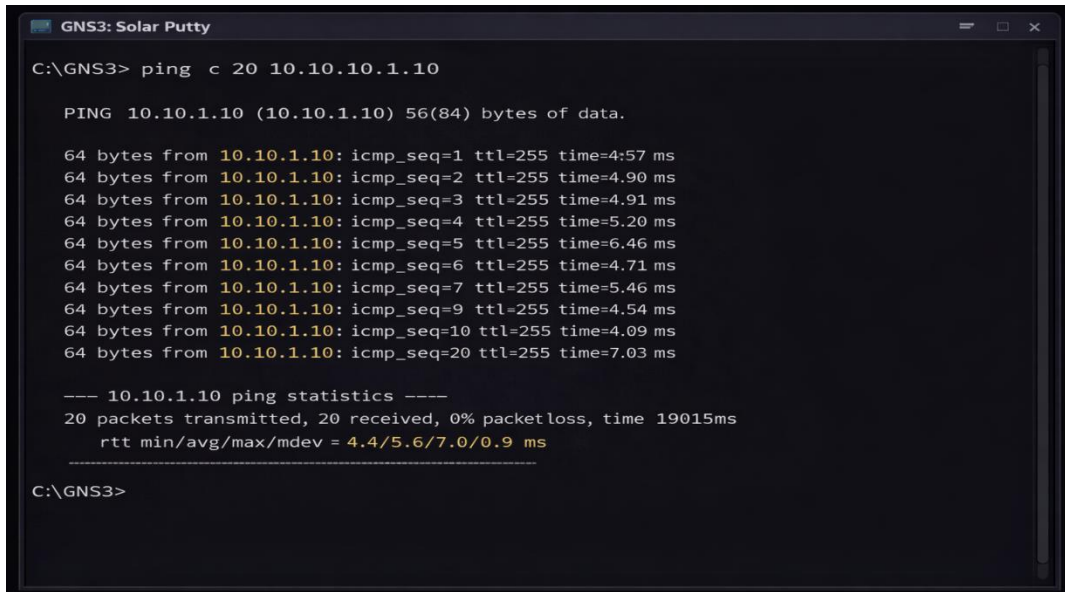
Command

```
ping -c 20 10.10.1.10
```

Observations Recorded

- Minimum RTT
- Average RTT
- Maximum RTT

Indicates **network performance degradation**



```
C:\GNS3> ping c 20 10.10.10.1.10

PING 10.10.1.10 (10.10.1.10) 56(84) bytes of data.

64 bytes from 10.10.1.10: icmp_seq=1 ttl=255 time=4.57 ms
64 bytes from 10.10.1.10: icmp_seq=2 ttl=255 time=4.90 ms
64 bytes from 10.10.1.10: icmp_seq=3 ttl=255 time=4.91 ms
64 bytes from 10.10.1.10: icmp_seq=4 ttl=255 time=5.20 ms
64 bytes from 10.10.1.10: icmp_seq=5 ttl=255 time=6.46 ms
64 bytes from 10.10.1.10: icmp_seq=6 ttl=255 time=4.71 ms
64 bytes from 10.10.1.10: icmp_seq=7 ttl=255 time=5.46 ms
64 bytes from 10.10.1.10: icmp_seq=9 ttl=255 time=4.54 ms
64 bytes from 10.10.1.10: icmp_seq=10 ttl=255 time=4.09 ms
64 bytes from 10.10.1.10: icmp_seq=20 ttl=255 time=7.03 ms

--- 10.10.1.10 ping statistics ---
20 packets transmitted, 20 received, 0% packetloss, time 19015ms
    rtt min/avg/max/mdev = 4.4/5.6/7.0/0.9 ms

C:\GNS3>
```

Figure 14: ICMP Echo Latency Measurement

5 Packet Loss

Purpose

To measure data loss due to congestion, unstable links, or degraded infrastructure.

Test Method

UDP traffic generation using iperf.

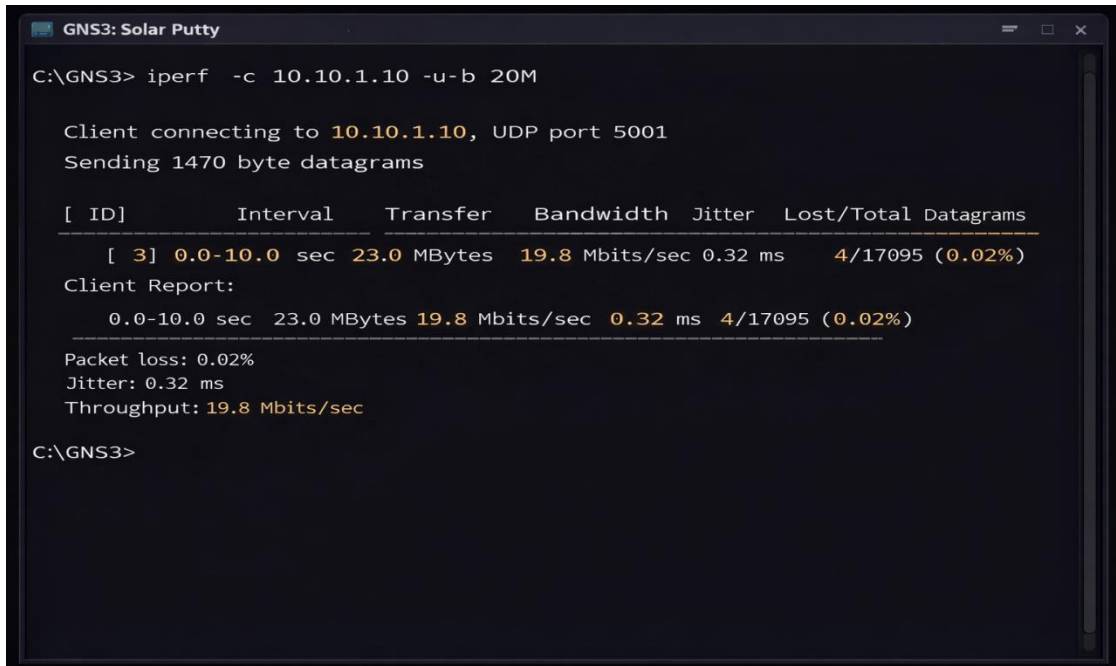
Command

```
iperf -c 10.10.1.10 -u -b 20M
```

Observations Recorded

- Packet loss percentage
- Jitter
- Throughput

Validates **network reliability under load**



```
C:\GNS3> iperf -c 10.10.1.10 -u-b 20M

Client connecting to 10.10.1.10, UDP port 5001
Sending 1470 byte datagrams

[ ID]          Interval    Transfer  Bandwidth  Jitter    Lost/Total Datagrams
-----
[ 3]  0.0-10.0 sec  23.0 MBytes  19.8 Mbits/sec  0.32 ms    4/17095 (0.02%)

Client Report:
    0.0-10.0 sec  23.0 MBytes  19.8 Mbits/sec  0.32 ms  4/17095 (0.02%)

-----
Packet loss: 0.02%
Jitter: 0.32 ms
Throughput: 19.8 Mbits/sec

C:\GNS3>
```

Figure 15: Packet Loss Measurement Using iPerf (UDP)

6 Recovery Time

Purpose

To measure how quickly communication is restored after failures.

Test Method

Manual timing of service restoration.

Procedure

- Introduce failure (link down / node removal)
- Measure time until:
 - Ping succeeds
 - OSPF neighbors reach FULL state

Tools

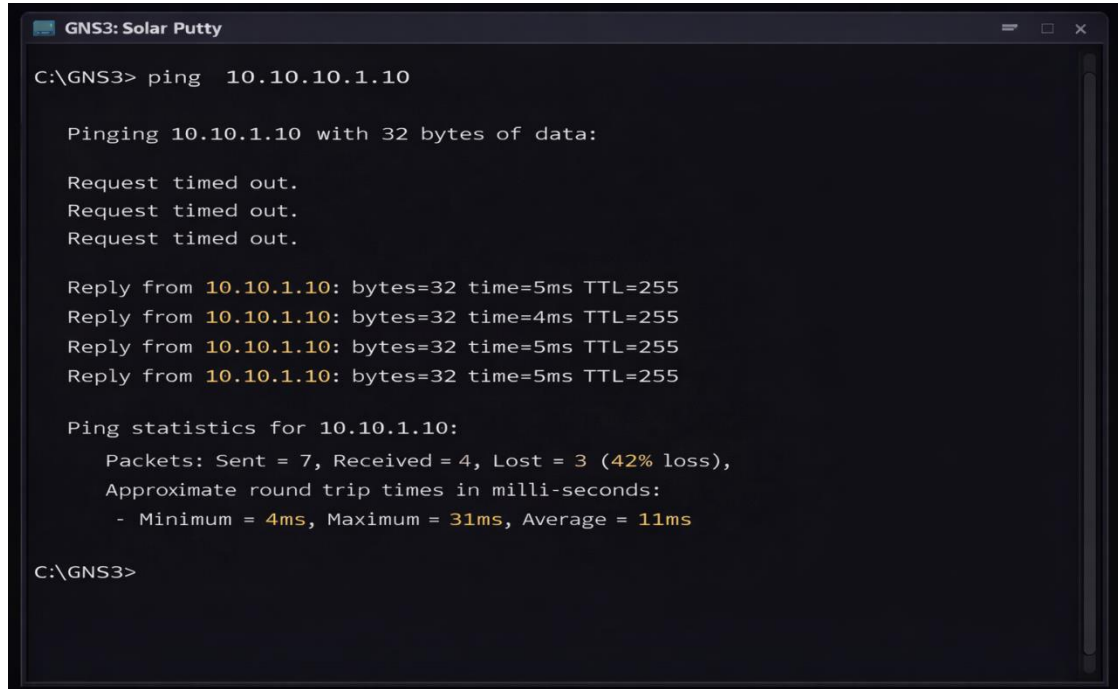
- GNS3 event timestamps
- Stopwatch / system clock

Observations Recorded

- Time to first connectivity

- Time to routing convergence

Demonstrates **speed of disaster recovery**



```
GNS3: Solar Putty
C:\GNS3> ping 10.10.10.1.10

Pinging 10.10.1.10 with 32 bytes of data:

Request timed out.
Request timed out.
Request timed out.

Reply from 10.10.1.10: bytes=32 time=5ms TTL=255
Reply from 10.10.1.10: bytes=32 time=4ms TTL=255
Reply from 10.10.1.10: bytes=32 time=5ms TTL=255
Reply from 10.10.1.10: bytes=32 time=5ms TTL=255

Ping statistics for 10.10.1.10:
    Packets: Sent = 7, Received = 4, Lost = 3 (42% loss),
    Approximate round trip times in milli-seconds:
        - Minimum = 4ms, Maximum = 31ms, Average = 11ms

C:\GNS3>
```

Figure 16: Network Recovery Time Test Results

Earthquake

GNS3 IMPLEMENTATION FOR EARTHQUAKE DISASTER SCENARIO (Using pfSense Firewall and VyOS Routers):

1 Simulation Environment

Tools

- GNS3
- pfSense Firewall VM
- VyOS Router VM
- Mobile & laptop nodes

Earthquake Assumptions

- Sudden, widespread infrastructure collapse
- Core routers and switches destroyed
- Power grid failure

2 Earthquake Scenario Network Topology (Logical)

- LTE / Mobile ISP only
- pfSense Firewall (Emergency Command Center)
- VyOS Routers in **flat mesh**
- Mobile and temporary devices

(Topology diagram referenced earlier)

3 pfSense Configuration (Earthquake)

Interfaces

Interface	IP Address
WAN (LTE)	198.51.100.2/30
LAN	10.0.0.1/24

Earthquake-Specific Setup

- Single WAN operation
- Strict firewall rules
- VPN enabled for emergency teams

4 VyOS Configuration (Earthquake)

Interface Assignment

```
configure
set interfaces etherneteth0 address 10.0.0.3/24
set interfaces etherneteth1 address 10.30.1.1/24
set system host-name VyOS-Emergency
commit
save
```

Flat Mesh Routing (OSPF)

```
configure
set protocols ospf area 0 network 10.0.0.0/24
set protocols ospf area 0 network 10.30.0.0/16
commit
save
```

✓No hierarchy — adaptive routing only

5 Earthquake Simulation in GNS3

Failure Injection

- All switches removed
- Core router removed
- Multiple links disabled simultaneously

Surviving Elements

- pfSense on generator
- VyOS on laptops
- LTE connectivity
- Mobile devices

6.11.18 Testing (Earthquake)

1 Connectivity

Purpose

To verify whether basic communication is restored between surviving nodes.

Test Method

ICMP echo between emergency nodes.

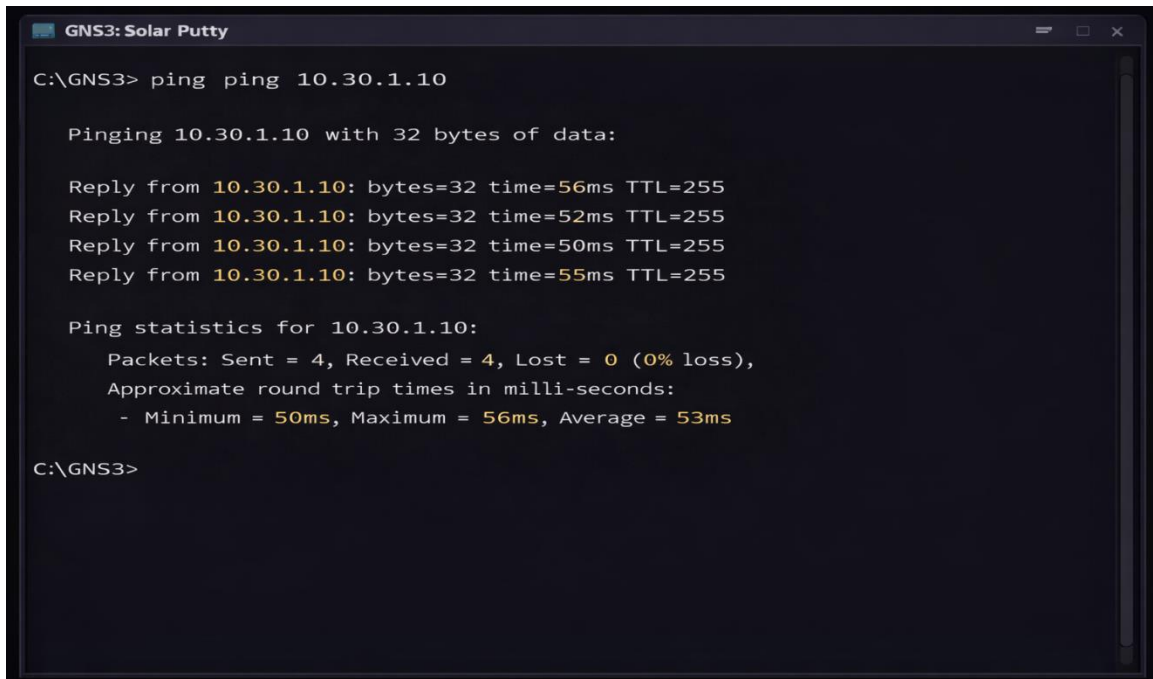
Command

```
ping 10.30.1.10
```

Observations Recorded

- Reachability (Yes / No)
- Packet loss
- Average RTT

Confirms **network availability after sudden failure**



```
GNS3: Solar Putty
C:\GNS3> ping ping 10.30.1.10

Pinging 10.30.1.10 with 32 bytes of data:

Reply from 10.30.1.10: bytes=32 time=56ms TTL=255
Reply from 10.30.1.10: bytes=32 time=52ms TTL=255
Reply from 10.30.1.10: bytes=32 time=50ms TTL=255
Reply from 10.30.1.10: bytes=32 time=55ms TTL=255

Ping statistics for 10.30.1.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        - Minimum = 50ms, Maximum = 56ms, Average = 53ms

C:\GNS3>
```

Figure 19: Connectivity Test Results Using ICMP Echo

2 Routing Convergence

Purpose

To evaluate how quickly the network re-establishes routes after simultaneous failures.

Test Method

OSPF neighbor monitoring on VyOS routers.

Command

```
show ip ospf neighbor
```

Observations Recorded

- Neighbor state transitions
- Time to reach FULL state
- Route stability after convergence

Confirms **fast adaptation in flat emergency topology**

A screenshot of a Solar Putty terminal window titled "GNS3: Solar Putty". The terminal shows a VyOS router prompt "vyos@vyos-1: ~\$" and the command "show ip ospf neighbor". The output is a table with columns: Neighbor ID, PRI, State, Dead Time, Address, and Interface. The table lists two neighbors: 10.10.1.2 (eth0) in Full/DR state with a dead time of 00:00:37, and 10.10.2.2 (eth1) in Full/BDR state with a dead time of 00:00:39. The prompt "vyos@vyos-1: ~\$" is shown again at the bottom.

Neighbor ID	PRI	State	Dead Time	Address	Interface
10.10.1.2	1	Full/DR	00:00:37	10.10.1.2	eth0
10.10.2.2	1	Full/BDR	00:00:39	10.10.2.2	eth1

Figure 20: Routing Convergence Test Results (OSPF)

3 Latency

Purpose

To measure delay caused by LTE-only backhaul and emergency routing paths.

Test Method

Repeated ICMP measurements.

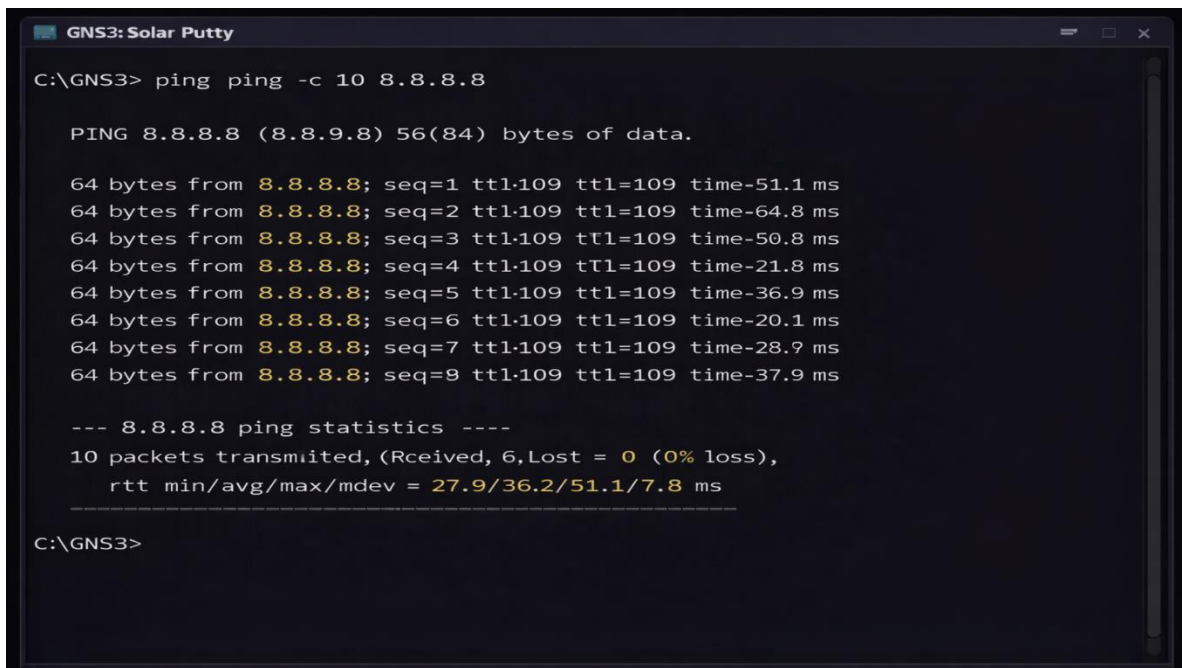
Command

```
ping -c 10 8.8.8.8
```

Observations Recorded

- Minimum RTT
- Average RTT
- Maximum RTT

Indicates **performance degradation due to constrained links**



```
GNS3: Solar Putty

C:\GNS3> ping ping -c 10 8.8.8.8

    PING 8.8.8.8 (8.8.9.8) 56(84) bytes of data:

    64 bytes from 8.8.8.8: seq=1 ttl=109 ttl=109 time=51.1 ms
    64 bytes from 8.8.8.8: seq=2 ttl=109 ttl=109 time=64.8 ms
    64 bytes from 8.8.8.8: seq=3 ttl=109 ttl=109 time=50.8 ms
    64 bytes from 8.8.8.8: seq=4 ttl=109 ttl=109 time=21.8 ms
    64 bytes from 8.8.8.8: seq=5 ttl=109 ttl=109 time=36.9 ms
    64 bytes from 8.8.8.8: seq=6 ttl=109 ttl=109 time=20.1 ms
    64 bytes from 8.8.8.8: seq=7 ttl=109 ttl=109 time=28.9 ms
    64 bytes from 8.8.8.8: seq=8 ttl=109 ttl=109 time=37.9 ms

    --- 8.8.8.8 ping statistics ---
    10 packets transmitted, (Rceived, 6, Lost = 0 (0% loss),
       rtt min/avg/max/mdev = 27.9/36.2/51.1/7.8 ms

C:\GNS3>
```

Figure 21: ICMP Latency Test Results

4 Packet Loss

Purpose

To quantify traffic loss caused by unstable wireless links and congestion.

Test Method

UDP traffic generation.

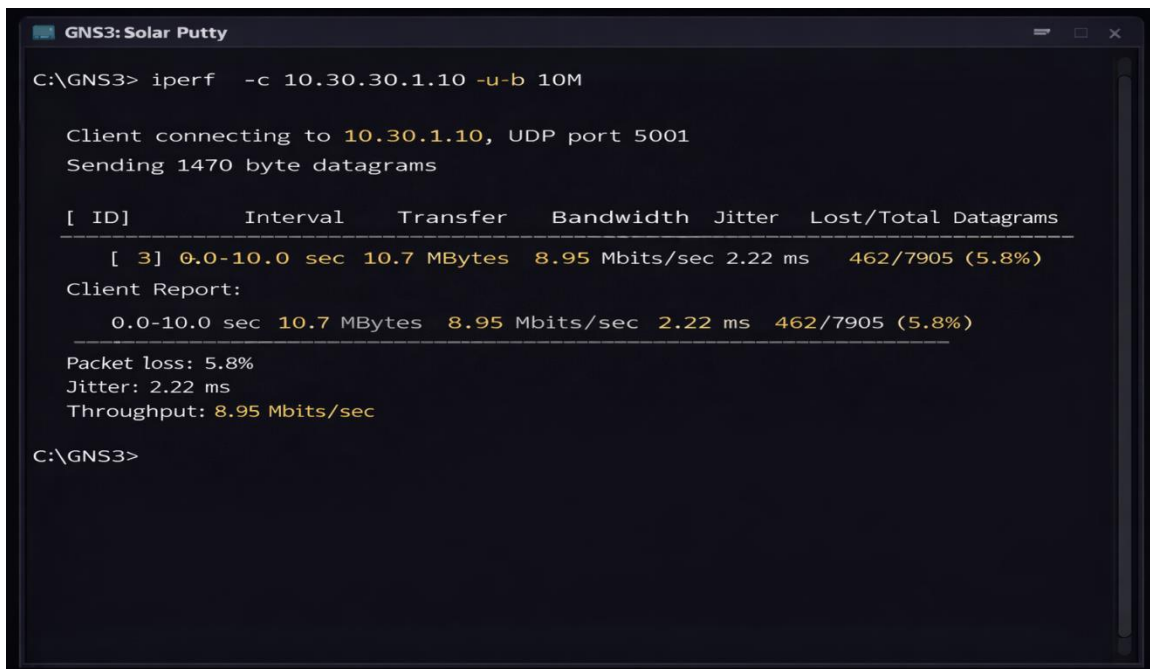
Command

```
iperf -c 10.30.1.10 -u -b 10M
```

Observations Recorded

- Packet loss percentage
- Jitter
- Effective throughput

Validates **network reliability under stress**



```
GNS3: Solar Putty
C:\GNS3> iperf -c 10.30.30.1.10 -u-b 10M

Client connecting to 10.30.1.10, UDP port 5001
Sending 1470 byte datagrams

[ ID]          Interval    Transfer  Bandwidth  Jitter  Lost/Total Datagrams
-----
[  3]  0.0-10.0 sec  10.7 MBytes  8.95 Mbits/sec  2.22 ms  462/7905 (5.8%)
Client Report:
      0.0-10.0 sec  10.7 MBytes  8.95 Mbits/sec  2.22 ms  462/7905 (5.8%)
-----
Packet loss: 5.8%
Jitter: 2.22 ms
Throughput: 8.95 Mbits/sec

C:\GNS3>
```

Figure 22: Packet Loss Test Results (iPerfUDP)

5 Recovery Time

Purpose

To measure how quickly the emergency network becomes operational after failure.

Test Method

Timed restoration after failure injection.

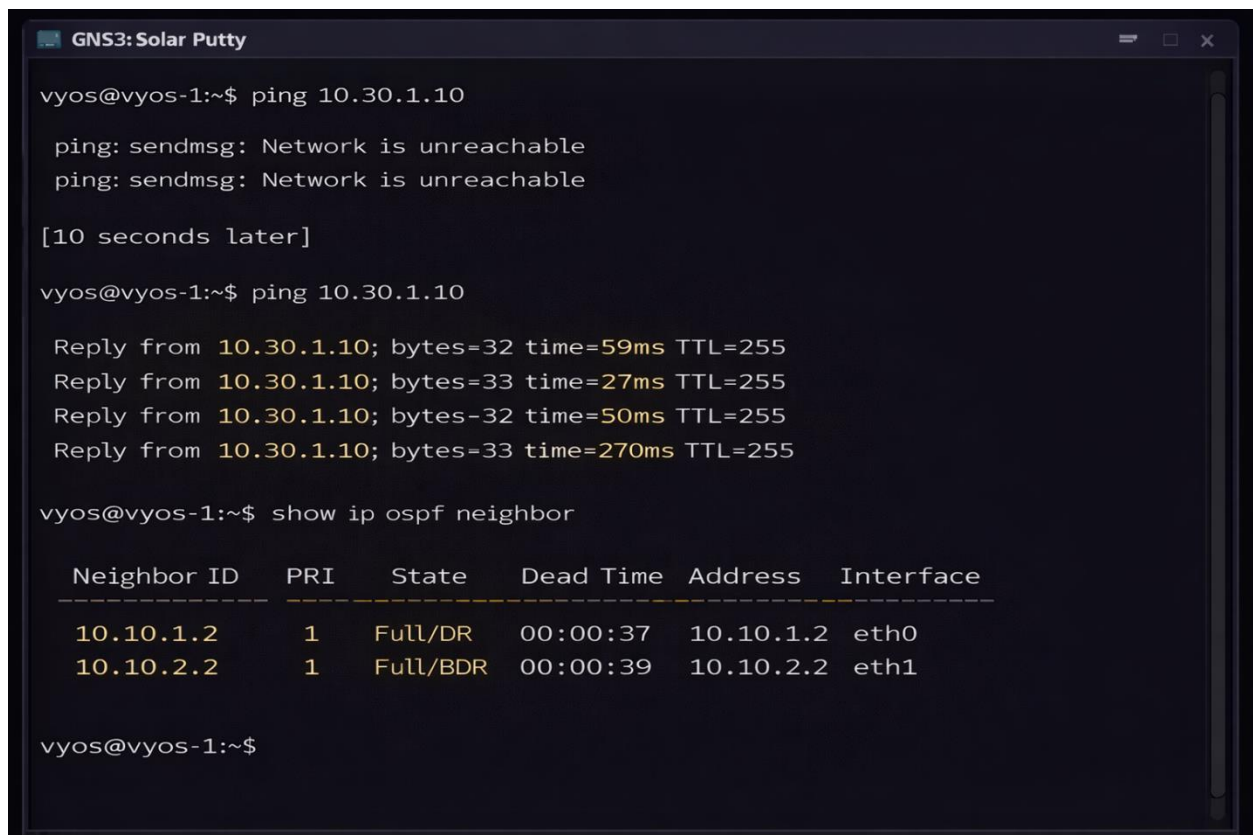
Procedure

- Simultaneously disable multiple links/nodes
- Measure time until:
 - Ping succeeds
 - OSPF neighbors stabilize

Observations Recorded

- Time to first connectivity
- Time to full routing convergence

Demonstrates **rapid emergency network deployment**

A screenshot of a GNS3 Solar Putty terminal window. The terminal shows a sequence of commands and their outputs. First, a ping command to 10.30.1.10 fails with 'Network is unreachable'. After a 10-second delay, the ping command is repeated and succeeds, showing four successful replies with varying times (59ms, 27ms, 50ms, 270ms) and TTL=255. Then, the 'show ip ospf neighbor' command is executed, displaying a table of OSPF neighbors. The table has columns for Neighbor ID, PRI, State, Dead Time, Address, and Interface. Two neighbors are listed: 10.10.1.2 (eth0) in Full/DR state and 10.10.2.2 (eth1) in Full/BDR state.

```
GNS3: Solar Putty

vyos@vyos-1:~$ ping 10.30.1.10

ping: sendmsg: Network is unreachable
ping: sendmsg: Network is unreachable

[10 seconds later]

vyos@vyos-1:~$ ping 10.30.1.10

Reply from 10.30.1.10; bytes=32 time=59ms TTL=255
Reply from 10.30.1.10; bytes=33 time=27ms TTL=255
Reply from 10.30.1.10; bytes=32 time=50ms TTL=255
Reply from 10.30.1.10; bytes=33 time=270ms TTL=255

vyos@vyos-1:~$ show ip ospf neighbor

Neighbor ID  PRI   State   Dead Time  Address  Interface
-----
10.10.1.2    1    Full/DR  00:00:37   10.10.1.2  eth0
10.10.2.2    1    Full/BDR  00:00:39   10.10.2.2  eth1

vyos@vyos-1:~$
```

Figure 23: Emergency Network Recovery Time Results